

Penetrating Injury to the Carotid Artery: Characterizing Presentation and Outcomes from the National Trauma Data Bank

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Background: Penetrating injury to the neck can be devastating because of the multiple vital structures in close proximity. In the event of injury to the carotid artery, there is a significantly increased likelihood of morbidity or mortality. The purpose of this study was to assess presenting characteristics associated with penetrating injury to the carotid artery and directly compare approaches to surgical management.

Methods: Data from the National Trauma Data Bank from 2002–2016 were accessed to evaluate adult patients sustaining penetrating injury to the common or internal carotid artery. Management (operative versus nonoperative) and surgical approach (open versus endovascular) were evaluated based on presentation characteristics, and outcomes were compared after propensity score matching.

Results: Three thousand three hundred ninety-one patients fitting inclusion criteria and surviving past the emergency department were included in analyses (nonoperative: 1,976 [58.3%] patients and operative: 1,415 [41.7%] patients). The operative group was further classified by intervention as open = 1,192 patients and endovascular: 154 patients. On presentation, the nonoperative group demonstrated significantly higher prevalence of coma (Glasgow Coma Scale ≤ 8 : nonoperative = 49.3% versus operative = 40.8%, $P < 0.001$), severe overall injury burden (Injury Severity Score ≥ 25 : nonoperative = 42.3% versus operative = 33.3%, $P < 0.001$), and severe head injury (Abbreviated Injury Score ≥ 3 : nonoperative = 44.9% versus operative = 22.0%, $P < 0.001$). After propensity score matching, the nonoperative group demonstrated higher mortality (nonoperative = 28.9% versus operative = 18.5%, $P < 0.001$), and lower rates of stroke (nonoperative = 6.6% versus operative = 10.5%, $P < 0.001$). There were no differences in outcomes relating to surgical approach.

Conclusions: These results indicate that nonoperative patients often present with a more severe overall injury burden, particularly injury to the head, and not surprisingly, have higher rates of mortality. The lack of significant differences in outcomes relating to surgical approach indicates open versus endovascular intervention should be individualized to the patient—for example, based on presenting characteristics and the location of the injury.

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BACKGROUND

Penetrating neck injury (PNI) encompasses up to 10% of all penetrating trauma and includes concomitant injury to the carotid artery for 6–20% of these patients.^{1–3} This population is particularly challenging because of the numerous vital structures within the neck often injured simultaneously, especially after trauma from high-velocity weapons.⁴ By extension, high rates of morbidity